

Teorema lui Peano:

Fie funcționala $\mathcal{L}: C^m[a,b] \rightarrow \mathbb{R}$ a.i. $\mathcal{L}(P) = 0$
 $\forall P \in \Pi_{m-1}$, ($\ker(\mathcal{L}) = \Pi_{m-1}$, i.e. se anulează pt orice
 polinom de grad mai mic sau egal cu $m-1$)

Atunci

$$\mathcal{L}(f) = \int_a^b k(t) f^{(m)}(t) dt$$

unde

$$k(t) = \frac{1}{(m-1)!} \mathcal{L}_* \left((x-t)_+^{m-1} \right) \quad \text{se numește}$$

nucleul lui Peano

Am notat $z_+ = \begin{cases} z, & z > 0 \\ 0, & z \leq 0 \end{cases}$

$\uparrow$ parte pozitivă

$$z_+^m = (z_+)^m$$

$\uparrow$ puterea trunchiată

Mai mult dacă k păstrează semn constant
 pe $[a,b]$ atunci $\exists \xi \in [a,b]$ a.i.

$$\mathcal{L}(f) = \frac{f^{(m)}(\xi)}{m!} \mathcal{L}(x^m)$$

În cazul nostru funcționalele au forma

MIDPOINT $R_d(f) = \int_a^b f(x) dx - (b-a) f\left(\frac{a+b}{2}\right)$

TRAPEZ $R_t(f) = \int_a^b f(x) dx - \frac{b-a}{2} (f(a) + f(b))$

SIMPSON $R_s(f) = \int_a^b f(x) dx - \frac{b-a}{6} (f(a) + 4f\left(\frac{a+b}{2}\right) + f(b))$

SIMPSON 3/8 $R_{3/8}(f) = \int_a^b f(x) dx - \frac{b-a}{8} \left(f(a) + 3f\left(\frac{2a+b}{3}\right) + \right.$
 $\left. + 3f\left(\frac{a+2b}{3}\right) + f(b) \right)$

$$= 9 =$$

$$R_d(f) = 0 \quad \forall \quad P \in \Pi_1 \quad (n-1=1 \Rightarrow n=2 \text{ pt teorema lui Peano})$$

$$\Rightarrow R_d(f) = \int_a^b K_1(t) f''(t) dt$$

unde

$$\begin{aligned} K_1(t) &= \frac{1}{(2-1)!} R_{d*}((x-t)_+^{2-1}) = R_{d*}((x-t)_+) = \\ &= \int_a^b (x-t)_+ dx - (b-a) \left(\frac{a+b}{2} - t \right)_+ = \\ &= \underbrace{\int_a^t (x-t)_+ dx}_{=0} + \int_t^b (x-t)_+ dx - (b-a) \left(\frac{a+b}{2} - t \right)_+ = \\ &= \int_t^b (x-t) dx - (b-a) \left(\frac{a+b}{2} - t \right)_+ = \begin{cases} \frac{(x-t)^2}{2} \Big|_t^b - (b-a) \left(\frac{a+b}{2} - t \right), & t \leq \frac{a+b}{2} \\ \frac{(x-t)^2}{2} \Big|_t^b, & t > \frac{a+b}{2} \end{cases} \\ &= \begin{cases} \frac{(b-t)^2}{2} - (b-a) \left(\frac{a+b}{2} - t \right), & t \leq \frac{a+b}{2} \\ \frac{(b-t)^2}{2}, & t > \frac{a+b}{2} \end{cases} = \begin{cases} \frac{(t-a)^2}{2}, & t \leq \frac{a+b}{2} \\ \frac{(b-t)^2}{2}, & t > \frac{a+b}{2} \end{cases} \end{aligned}$$

$$\Rightarrow K_1(t) = \begin{cases} \frac{(t-a)^2}{2}, & t \leq \frac{a+b}{2} \\ \frac{(b-t)^2}{2}, & t > \frac{a+b}{2} \end{cases} \Rightarrow K_1(t) \geq 0 \quad \forall t \in [a, b]$$

$\Rightarrow K_1$ păstrează semn constant pe $[a, b] \Rightarrow \exists \zeta \in [a, b]$

a.i. $R_d(f) = \frac{f''(\zeta)}{2!} R_d(x^2)$. Am văzut că

$$R_d(x^2) = \frac{b^3-a^3}{3} - (b-a) \left(\frac{a+b}{2} \right)^2 = (b-a) \left(\frac{a^2+ab+b^2}{3} - \right.$$

$$\left. \frac{a^2+2ab+b^2}{4} \right) = \int_a^b x^2 dx$$

$$= \frac{b-a}{12} (4a^2 + 4ab + 4b^2 - 3a^2 - 6ab - 3b^2) =$$

$$= \frac{b-a}{12} (a^2 - 2ab + b^2) = \frac{(b-a)^3}{12}$$

$$\Rightarrow R_d(f) = \frac{(b-a)^3}{24} f''(\zeta)$$

Pt a calcula K_1 s-a folosit

$$\left(\frac{a+b}{2} - t \right)_+ = \begin{cases} \frac{a+b}{2} - t, & t \leq \frac{a+b}{2} \\ 0, & t > \frac{a+b}{2} \end{cases}$$

$$R_1(P) = 0 \quad \forall P \in \Pi_1 \quad (m-1=1 \Rightarrow m=2 \text{ pt teorema lui Peano})$$

$$\Rightarrow R_1(f) = \int_a^b K_2(t) f''(t) dt$$

unde

$$K_2(t) = \frac{1}{(2-1)!} R_{1x}((x-t)_+^{2-1}) = R_{1x}((x-t)_+) =$$

$$= \int_a^b (x-t)_+ dx - \frac{b-a}{2} [(a-t)_+ + (b-t)_+] =$$

$$= \underbrace{\int_a^t (x-t)_+ dx}_{=0} + \int_t^b (x-t)_+ dx - \frac{b-a}{2} [\underbrace{(a-t)_+}_{=0} + \underbrace{(b-t)_+}_{=b-t}] =$$

$$= \int_t^b (x-t) dx - \frac{b-a}{2} (b-t) = \frac{(x-t)^2}{2} \Big|_t^b - \frac{(b-a)(b-t)}{2} =$$

$$= \frac{(b-t)^2}{2} - \frac{(b-a)(b-t)}{2} = \frac{(b-t)(a-t)}{2} \Rightarrow K_2(t) \leq 0 \quad \forall t \in [a, b]$$

$\Rightarrow K_2$ păstrează semn constant pe $[a, b] \Rightarrow \exists \zeta \in [a, b]$

a.i. $R_1(f) = \frac{f''(\zeta)}{2!} R_1(x^2)$. Am văzut că

$$R_1(x^2) = \frac{b^3 - a^3}{3} - \frac{b^3 - ab^2 + a^2b - a^3}{2} = \frac{-b^3 + 3ab^2 - 3a^2b + a^3}{6} =$$

$$= -\frac{(b-a)^3}{6}$$

$$\Rightarrow R_1(f) = \frac{-(b-a)^3}{6} \frac{f''(\zeta)}{2}$$

$$R_3(P) = 0 \quad \forall P \in \Pi_3 \quad (m-1=3 \Rightarrow m=4 \text{ pt teorema lui Peano})$$

$$\Rightarrow R_3(f) = \int_a^b K_4(t) f^{(4)}(t) dt$$

unde

$$K_4(t) = \frac{1}{(4-1)!} R_{3x}((x-t)_+^{4-1}) = \frac{1}{3!} R_{3x}((x-t)_+^3)$$

$$= 11 =$$

$$= \frac{1}{3!} \int_a^b (x-t)_+^3 dx - \frac{b-a}{6} \left[(a-t)_+^3 + 4 \left(\frac{a+b}{2} - t \right)_+^3 + (b-t)_+^3 \right] \frac{1}{3!} =$$

$$= \frac{1}{3!} \int_a^t \underbrace{(x-t)_+^3}_{=0} dx + \frac{1}{3!} \int_t^b \underbrace{(x-t)_+^3}_{=(x-t)^3} dx - \frac{b-a}{6} \left[\underbrace{(a-t)_+^3}_{=0} + 4 \left(\frac{a+b}{2} - t \right)_+^3 + (b-t)_+^3 \right] \frac{1}{3!} = \frac{1}{3!} \int_t^b (x-t)^3 dx - \frac{1}{3!} \frac{b-a}{6} \left[4 \left(\frac{a+b}{2} - t \right)_+^3 + (b-t)^3 \right] + (b-t)^3 =$$

$$= \frac{(b-t)^4}{4 \cdot 3!} - \frac{b-a}{6} \frac{1}{3!} \left[4 \left(\frac{a+b}{2} - t \right)_+^3 + (b-t)^3 \right] =$$

$$= \frac{1}{3!} \begin{cases} \frac{(b-t)^4}{4} - \frac{b-a}{6} \left[4 \left(\frac{a+b}{2} - t \right)^3 + (b-t)^3 \right], & a \leq t \leq \frac{a+b}{2} \\ \frac{(b-t)^4}{4} - \frac{b-a}{6} (b-t)^3, & \frac{a+b}{2} < t \leq b \end{cases}$$

De exemplu

$$K_3(t) = \frac{1}{3!} \frac{(b-t)^4}{4} - \frac{1}{3!} \frac{b-a}{6} (b-t)^3 = \frac{1}{3!} \frac{(b-t)^3}{2} \left(\frac{b-t}{2} - \frac{b-a}{3} \right) =$$

$$= \frac{1}{3!} \frac{(b-t)^3}{2} \frac{1}{6} (3b-3t-2b+2a) = \frac{1}{3!} \frac{(b-t)^3}{12} (b+2a-3t)$$

pentru $\frac{a+b}{2} < t \leq b$

$$\text{Din } \frac{a+b}{2} < t \mid (-3) \Rightarrow -\frac{3}{2}a - \frac{3}{2}b > -3t \mid +(b+2a)$$

$$-\frac{b}{2} + \frac{a}{2} > -3t + b + 2a \Rightarrow -\frac{b-a}{2} > b+2a-3t$$

$$\Rightarrow K_3(t) \leq 0 \quad \forall \frac{a+b}{2} < t \leq b$$

Analog se arată că $K_3(t) \leq 0 \quad \forall \frac{a+b}{2} > t \geq a$

$\Rightarrow K_3$ păstrează semn constant pe $[a, b] \Rightarrow \exists \eta \in [a, b]$
 a.i. $R_5(f) = \frac{f^{(4)}(\eta)}{4!} R_5(x^4)$

Am văzut că $=12=$

$$R_5(x^4) = \frac{b^5 - a^5}{5} - \frac{b-a}{24} (5a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 5b^4) =$$

$$= -\frac{b-a}{30} (a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4) = -\frac{(b-a)^5}{90}$$

$$\boxed{R_5(f) = \frac{-(b-a)^5}{2880} f^{(4)}\left(\frac{1}{4}\right)}$$

$R_{3/8}(f) = 0 \quad \forall f \in \Pi_3$ ($n-1=3 \Rightarrow n=4$ teorema lui Peano)

$$\Rightarrow R_{3/8}(f) = \int_a^b K_4(t) f^{(4)}(t) dt$$

unde

$$K_4(t) = \frac{1}{(4-1)!} R_{3/8}((x-t)_+^{4-1}) = \frac{1}{3!} R_{3/8}((x-t)_+^3) =$$

$$= \frac{1}{3!} \int_a^b (x-t)_+^3 dx - \frac{b-a}{8} \left[(a-t)_+^3 + 3\left(\frac{2a+b}{3}-t\right)_+^3 + \right.$$

$$\left. + 3\left(\frac{a+2b}{3}-t\right)_+^3 + (b-t)_+^3 \right] \frac{1}{3!} =$$

$$= \frac{1}{3!} \int_a^t (x-t)_+^3 dx + \frac{1}{3!} \int_t^b (x-t)_+^3 dx - \frac{b-a}{8} \frac{1}{3!} \left[\underbrace{(a-t)_+^3}_{=0} + \right.$$

$$\left. + 3\left(\frac{2a+b}{3}-t\right)_+^3 + 3\left(\frac{a+2b}{3}-t\right)_+^3 + \underbrace{(b-t)_+^3}_{=(b-t)^3} \right] =$$

$$= \frac{1}{3!} \int_t^b (x-t)^3 dx - \frac{1}{3!} \frac{b-a}{8} \left[3\left(\frac{2a+b}{3}-t\right)_+^3 + 3\left(\frac{a+2b}{3}-t\right)_+^3 + \right.$$

$$\left. + (b-t)^3 \right] = \frac{1}{3!} \frac{(x-t)^4}{4} \Big|_t^b - \frac{1}{3!} \frac{b-a}{8} \left[3\left(\frac{2a+b}{3}-t\right)_+^3 + 3\left(\frac{a+2b}{3}-t\right)_+^3 + \right.$$

$$\left. + (b-t)^3 \right] = \begin{cases} 3\left(\frac{2a+b}{3}-t\right)^3 + 3\left(\frac{a+2b}{3}-t\right)^3 + (b-t)^3, & a \leq t \leq \frac{2a+b}{3} \\ 3\left(\frac{a+2b}{3}-t\right)^3 + (b-t)^3, & \frac{2a+b}{3} < t \leq \frac{a+2b}{3} \\ (b-t)^3, & \frac{a+2b}{3} < t \leq b \end{cases}$$

$$= \frac{(b-t)^4}{24} - \frac{b-a}{48} \begin{cases} 3\left(\frac{2a+b}{3}-t\right)^3 + 3\left(\frac{a+2b}{3}-t\right)^3 + (b-t)^3, & \frac{2a+b}{3} < t \leq \frac{a+2b}{3} \\ (b-t)^3, & \frac{a+2b}{3} < t \leq b \end{cases}$$

$$= 1/3 =$$

De exemplu

$$K_4(t) = \frac{(b-t)^4}{24} - \frac{b-a}{48} (b-t)^3 = \frac{(b-t)^3}{24} (b-t - 2b + 2a) =$$

$$= \frac{(b-t)^3}{48} (2a - b - t) \text{ pentru } \frac{a+2b}{3} < t \leq b$$

$$\frac{a+2b}{3} < t \quad | \cdot (-1) \Leftrightarrow -\frac{a}{3} - \frac{2b}{3} > -t \quad | + (2a-b)$$

$$\Leftrightarrow \frac{5a}{3} - \frac{5b}{3} > 2a - b - t \Leftrightarrow -\frac{5(b-a)}{3} > 2a - b - t$$

$$\Rightarrow K_4(t) \leq 0 \quad \forall \frac{a+2b}{3} < t \leq b$$

Analog se arată că $K_4(t) \leq 0 \quad \forall t \in [a, b]$

$\Rightarrow K_4$ păstrează semn constant pe $[a, b] \Rightarrow \exists \xi \in (a, b)$
c.î.

$$R_{3/8}(f) = \frac{f^{(4)}(\xi)}{4!} R_{3/8}(x^4)$$

Am văzut că

$$R_{3/8}(x^4) = \frac{b^5 - a^5}{5} - \frac{b-a}{8 \cdot 27} (44a^4 + 40a^3b + 48a^2b^2 + 40ab^3 + 44b^4)$$

$$= -\frac{b-a}{270} (a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4) =$$

$$= \frac{-(b-a)}{270} (b-a)^4 = -\frac{(b-a)^5}{270}$$

$$\Rightarrow \boxed{R_{3/8}(f) = \frac{-(b-a)^5}{6480} f^{(4)}(\xi)}$$